



Research Diary

Hui

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1 March 30, 2024, Manifolds with positive spectrum

1.1 Abstract

I'm preparing the for seminar in Fudan the chapter Manifolds with positive spectrum.

It needs operator theory especially for Unbounded operators, since we are only interesting in the Laplace operator. On the other hand, it requires algebraic topology theory for surfaces (may have boundary or punctured).

1.2 Interesting Papers

1. Bernard Helffer, *Spectral Theory and its Applications*, 2013

2. Yosida, *Functional analysis*.
3. Rudin, *Functional analysis*.
4. Simon004, *Operator theory*
5. Simon and Reed, *Methods of modern mathematical physics*
6. S H Kulkarni, M T Nair And G Ramesh, *Some properties of unbounded operators with closed range*
7. Friedrich Haslinger's notes, [https://www.mat.univie.ac.at/ has/](https://www.mat.univie.ac.at/has/)
8. Jean-Pierre Francoise, *Encyclopedia Of Mathematical Physics 12, Functional Analysis And Operator Algebras; Quantization Methods And Path Integration; Variational Techniques*, Science Press
9. Carlos S. Kubrusly, *The Elements of Operator Theory*
10. Cheng, *Eigenvalue comparison theorems and its geometric applications*
11. Chavel Isaac, *Eigenvalues in Riemannian geometry*

On surfaces theory, I refer

- [open learn](#)
- do Carmo, *Differential geometry of curves and surfaces*
- Jonathan L. Gross, Jay Yellen and Mark Anderson, *Graph Theory and Its Applications*, 2019

1.3 Mathematics

Definition 1. A **subdivision** of a surface consists of a finite set of points, called vertices, and a finite set of edges, as defined above, such that:

- each vertex is an endpoint of at least one edge;
- each endpoint of an edge is a vertex;
- the vertices and edges form a connected graph;
- no two edges have any points in common;
- if the surface has a boundary, the boundary consists only of vertices and edges;
- the space obtained from the surface by removing the vertices and edges is a union of disjoint pieces, called faces, each of which is homeomorphic to an open disc.

An example of a subdivision is the *cube subdivision* of a sphere, shown in Figure 1, which has 8 vertices, 12 edges and 6 faces.

Theorem 1. Inserting k holes into a surface reduces its Euler characteristic by k .

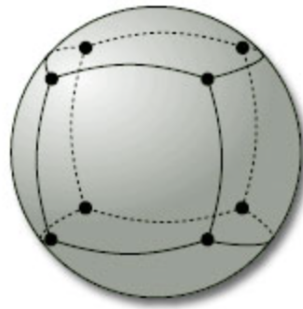


Figure 1: The cube subdivision of a sphere

1.4 Problems

1. What's the eigenvalues of the balls in a space form?

2 April 5, Bejancu's *Geometry of CR-Submanifolds*

2.1 Abstract

I'm reading Bejancu's *Geometry of CR-Submanifolds* and before that I was reading Blair's *Riemannian Geometry of Contact and Symplectic Manifolds*. Next step, I should go back to *foliation theory*.

2.2 Interesting Papers

- Yano and Kon, *CR Submanifolds of Kaehlerian and Sasakian Manifolds*
- Yano and Kon, *Anti-invariant submanifolds*
- Yano, *On a Structure Defined by a Tensor Field f of Type $(1,1)$ Satisfying $f^3 + f = 0$.*
- Yano, *Integral formulas in Riemannian geometry*
- Ogiue, *Differential geometry of Kaehler submanifolds*

2.3 Mathematics

"These two classes of submanifolds have been extensively investigated in the last decade from different viewpoints. The fundamental results on the geometry of totally real submanifolds can be found in Yano-Kon [1]. Also, a survey of the principal results on the geometry of complex submanifolds is given by Ogiue in [1]."

Yano-Kon's *Anti-invariant submanifolds* is not easy to find online, so I borrow it from UniVie. But you can find it in archive.org.

Bejancu's book is great, but it seems that there is a few misquoted or error. I mainly read the part of second chapter about CR submfd of an almost Hermitian manifold, and left the CR submfd in Kahler mfd to read when necessary. And the first chapter about basic theory of Rie mfd are also uesful.

3 April 6, Stability of minimal hypersurfaces in a 3-manifold

3.1 Abstract

- Joining Fudan's seminar.

3.2 Papers

1. Chen Zhou, *A new proof of Huber's theorem on differential geometry in the large*
2. Peter Li and Luen-fai Tam, *Complete Surfaces With Finite Total Curvature*

3.3 Mathematics

3.3.1 Huber's Theorem

Theorem 2 (Huber). *Let M be a connected open Riemann surface and let ds^2 be a complete conformal Riemannian metric (This metric can be expressed as (2) locally.) of class C^∞ on M . If the integral*

$$\int_M K^- d\sigma \quad (1)$$

is finite, then M is homeomorphic to the interior of a compact surface with boundary.

In order to understand the Huber's theorem, here I quote some preliminaries.

We consider an open, two-dimensional Riemannian manifold M whose metric is defined by a positive definite quadratic form

$$ds^2 = E(\xi, \eta)d\xi^2 + 2F(\xi, \eta)d\xi d\eta + G(\xi, \eta)d\eta^2,$$

where ξ and η denoting local parameters. If E , F and G are sufficiently regular, then it is possible to introduce (local) *isothermic parameters*, i.e., there exists a coordinate transformation $x = x(\xi, \eta)$, $y = y(\xi, \eta)$ such that $E = G > 0$, $F = 0$ in the (x, y) -parameter system. Then we can write

$$ds^2 = e^{2u(x,y)}(dx^2 + dy^2) = e^{2u}|dz|^2, \quad (2)$$

putting $z = x + iy$. Such a transformation always exists, for example, when E , F and G are of class C^3 , and in this case the corresponding function u is also of class C^3 (cf. A. Wintner, p. 687).

The well-known theorems of Cohn-Vossen [2] and Huber [8] assert that:

Proposition 1. *Let M be a complete surface with the negative part of its Gaussian curvature integrable, i.e.,*

$$\int_M K_- dA < \infty,$$

where

$$K_- = \begin{cases} 0 & \text{if } K > 0, \\ -K & \text{if } K \leq 0. \end{cases}$$

Then M must be conformally equivalent to a compact Riemann surface with finitely many points deleted. Moreover

$$\int_M K \, dA \leq 2\pi\chi(M),$$

where $\chi(M)$ is the Euler characteristic of M . In particular, if $\int_M |K| \, dA < \infty$, if in addition M is simply connected, then M must be conformally equivalent to the complex plane.

The following proposition was proved by Hartman in [6] for simply connected surfaces. It was later generalized by Shiohama in [15] to arbitrary complete surfaces with finite total curvature.

Proposition 2. *Let M be a complete surface with finite total curvature. If K is the Gaussian curvature of M , then we have*

$$2\pi\chi(M) - \int_M K \, dA = \lim_{r \rightarrow \infty} \frac{L(r)}{r} = \lim_{r \rightarrow \infty} \frac{2A(r)}{r^2}.$$

The next result, which was proved by Hartman in [6], asserts an upper bound of the area growth of a complete surface with finite total curvature. For a higher-dimensional generalization, we would like to refer the reader to [13].

Proposition 3. *Let M be a complete surface with finite total curvature. Then there exists a constant C_1 , depending only on $\int_M |K| \, dA$, such that*

$$L(\partial B_x(r)) \leq C_1 r \quad \text{and} \quad A(B_x(r)) \leq \frac{C_1}{2} r^2,$$

for all $x \in M$ and for all $r > 0$.

Proposition 4. *Let M be a simply connected complete noncompact surface with finite total curvature. Then there exists $R_0 > 0$, such that for all $r > R_0$, the boundary of the geodesic ball of radius r centered at p must be homeomorphic to the circle. In particular, the geodesic ball $B_p(r)$ is homeomorphic to the disk.*

3.3.2 The Uniformization Theorem

Theorem 3 (The Riemann Mapping Theorem or The Uniformization Theorem). *Suppose X is a simply connected Riemann surface. Then X can be mapped biholomorphically onto either the Riemann sphere $\mathbb{P}^1$, the complex plane $\mathbb{C}$ or else the unit disk D .*

Definition 2. Suppose X is a Riemann surface and $\tilde{X}$ is its universal covering. Since $\tilde{X}$ is simply connected, we may apply the Riemann Mapping Theorem to it. It is standard to call X **elliptic**, **parabolic** or **hyperbolic** depending on whether its universal covering is isomorphic to $\mathbb{P}^1$, $\mathbb{C}$, or D .

Theorem 4.

- (a) *The Riemann sphere $\mathbb{P}^1$ is elliptic.*
- (b) *The complex plane $\mathbb{C}$, the punctured plane $\mathbb{C}^*$ and all tori $\mathbb{C}/\Gamma$ are parabolic.*
- (c) *Every other Riemann surface is hyperbolic.*

3.3.3 Fischer-Colbrie–Schoen Theorem

In Li's book, the Fischer-Colbrie–Schoen Theorem is not clear. Here is the original paper's theorem.

Theorem 5 (Fischer-Colbrie–Schoen). *Let N be a complete oriented 3-manifold of non-negative scalar curvature. Let M be an oriented complete stable minimal surface in N .*

There are two possibilities:

(i) If M is compact, then M is conformally equivalent to the sphere S^2 or M is a totally geodesic flat torus T^2 . If $S > 0$ on N , then M is conformally equivalent to S^2 .

(ii) If M is not compact, then M is conformally equivalent to the complex plane $\mathbb{C}$, or the cylinder A . If M is a cylinder and the absolute total curvature of M is finite, then M is flat and totally geodesic. If the scalar curvature of N is everywhere positive, then M cannot be a cylinder with finite total curvature. If the Ricci curvature of N is non-negative, then the assumption of finite total curvature in (ii) can be removed.

Remark. Note that each of the four possibilities does occur.

4 April 7, Stokes Theorem

4.1 Abstraction

Covering maps, nonorientable manifolds, singular boundary.

4.2 Papers

- [Mathematics Magazine](#)

4.3 Mathematics

Lee's definition implies that the tangent space at a boundary point is an n -dimensional vector space, just like the tangent spaces at interior points.

Lemma 1. *Let $\tau: \mathbb{H}^n \rightarrow \mathbb{R}^n$ denote the inclusion map. For any $a \in \partial\mathbb{H}^n$, the differential $d\tau_a: T_a\mathbb{H}^n \rightarrow T_a\mathbb{R}^n$ is an isomorphism.*

5 April 10, Distance functions

5.1 Mathematics

Theorem 6. *Suppose (M, g) is a connected Riemannian manifold, $S \subseteq M$ is arbitrary, and $f: M \rightarrow [0, \infty)$ is the distance to S , that is, $f(x) = d_g(x, S)$ for all $x \in M$. If f is continuously differentiable on some open subset $U \subseteq M \setminus S$, then $|\nabla f| = 1$ on U .*

Theorem 7. *Suppose (M, g) is a Riemannian manifold and f is a smooth local distance function on an open subset $U \subseteq M$. Then $\nabla_{\nabla f}(\nabla f) = 0$, and each integral curve of ∇f is a unit-speed geodesic.*

Exercise 1. Suppose (M, g) is a complete, connected Riemannian manifold, $S \subseteq M$ is a closed subset, and $f: M \rightarrow \mathbb{R}$ is a continuous function such that $f^{-1}(0) = S$ and f is smooth with unit gradient on $M \setminus S$. Prove that $f(x) = d_g(x, S)$ for all $x \in M$.

Exercise 2. Suppose (M, g) is a Riemannian n -manifold, $P \subseteq M$ is a compact embedded submanifold, and U is an ϵ -tubular neighborhood of P for some $\epsilon > 0$. For $0 < \delta < \epsilon$, define U_δ and P_δ by

$$U_\delta = \{x \in U : d_g(x, P) \leq \delta\},$$

$$P_\delta = \{x \in U : d_g(x, P) = \delta\}.$$

(a) Prove that U_δ is a regular domain and P_δ is a compact, embedded submanifold of M for each $\delta \in (0, \epsilon)$.

(b) Generalize the result of Problem 6-30 by proving that

$$\text{Vol}(U_\delta) = \int_0^\delta \text{Area}(P_r) dr,$$

where $\text{Area}(P_r)$ denotes the $(n-1)$ -dimensional volume of P_r with the induced Riemannian metric.

Also in the book *Riemannian Foliations* by Molino, (see proposition 3.5)

“B.Reinhart established some important properties of these foliations and, in particular, the fact that every geodesic [for a bundle-like metric] that is perpendicular at some point to a leaf remains perpendicular to all the leaves that it meets. He also showed that if the metric is complete then the leaves have the same universal covering.”

LH 1. From above, we know that it's possible to define a distance function r to a compact section of a leaf $\mathcal{F}_{p_0}$ in a Riemannian foliation M . Then ∇r is unit-speed geodesic and thus horizontal. When M has focal points wrt $\mathcal{F}_{p_0}$, the integral on the whole space will not be affected, which gives a way to derive a monotonicity formula in general Riemannian foliation setting.

6 April 13, Stability of minimal hypersurfaces in a higher dimensional manifold

6.1 Papers

- [stackexchange](#)
- Cheeger, J., and Gromoll, D. (1972). *On the Structure of Complete Manifolds of Nonnegative Curvature*. Annals of Mathematics, 96(3), 413–443. <https://doi.org/10.2307/1970819>
- <https://www.ams.org/books/gsm/019/>
- [A solution of Carmo's book](#)
- [Steele Prize For Mathematical Exposition](#)

6.2 Mathematics

Exercise 3 (Existence of a ray, from Carmo's book). A geodesic $\gamma : [0, \infty) \rightarrow M$ in a Riemannian manifold M is called a ray starting from $\gamma(0)$ if it minimizes the distance between $\gamma(0)$ and $\gamma(s)$, for any $s \in (0, \infty)$. Assume that M is complete, non-compact, and let $p \in M$. Show that M contains a ray starting from p .

Corollary 1. Let E be an end of a complete Riemannian manifold. Suppose that for some $\nu > 1$ and $C > 0$, E satisfies a Sobolev type inequality of the form

$$\left(\int_E |u|^{2\nu} \right)^{\frac{1}{\nu}} \leq C \int_E |\nabla u|^2,$$

for all compactly supported functions $u \in H_c^{1,2}(E)$ defined on E , then E must be nonparabolic.

We know that $\mathbb{R}^2$ is parabolic, but $\mathbb{R}^n$ with $n \geq 3$ is nonparabolic. Compare to (note that the dimension works here)

Theorem 8 (Gagliardo–Nirenberg–Sobolev inequality). Assume $1 \leq p < n$. There exists a constant C , depending only on p and n , such that

$$\|u\|_{L^{p^*}(\mathbb{R}^n)} \leq C \|\nabla u\|_{L^p(\mathbb{R}^n)}, \quad (3)$$

for all $u \in C_c^1(\mathbb{R}^n)$.

6.2.1 About properly immersed submanifolds

Definition 3. An embedded submanifold with or without boundary is said to be *properly embedded* if the inclusion map is proper.

Proposition 5. If $\widetilde{M}$ is a smooth manifold with or without boundary, then an embedded submanifold $M \subseteq \widetilde{M}$ is properly embedded if and only if it is a closed subset of $\widetilde{M}$.

Remark (cf. [Shun Maeta](#)). An immersed submanifold M in a Riemannian manifold N is said to be properly immersed if the immersion is a proper map. Here, we remark that the properness of the immersion implies the completeness of (M, g) .

Proposition 6. Suppose X and Y are topological spaces and $F : X \rightarrow Y$ is a proper map. Then F takes every sequence diverging to infinity in X to a sequence diverging to infinity in Y .

Proposition 7. Suppose X is any topological space, Y is a compactly generated Hausdorff space (e.g., any subset of a manifold with or without boundary), and $F : X \rightarrow Y$ is a proper continuous map. Then F is a closed map.

Proposition 8. First countable spaces and locally compact spaces are compactly generated.

Proposition 9. Suppose F is a proper continuous map from a topological space to a compactly generated Hausdorff space.

- (a) If F is surjective, it is a quotient map.
- (b) If F is injective, it is a topological embedding.
- (c) If F is bijective, it is a homeomorphism.

Proposition 10. Suppose M is a smooth manifold with or without boundary, and $S \subseteq M$ is an immersed submanifold. If any of the following holds, then S is embedded.

1. S has codimension 0 in M .
2. The inclusion map $S \subseteq M$ is proper.
3. S is compact.

7 April 15, Tensor product of distributions

7.1 Papers

- Michael's note, *Theory of distributions*
- Rudin, *Functional analysis*
- [arxiv](#), It confirms a conjecture that all symmetric spaces of noncompact type posses complex values harmonic morphisms.

7.2 Mathematics

7.2.1 Types of topological vector spaces

In the following definitions, X always denotes a topological vector space, with topology τ .

1. X is *locally convex* if there is a local base $\mathcal{B}$ whose members are convex.
2. X is *locally bounded* if 0 has a bounded neighborhood.
3. X is *locally compact* if 0 has a neighborhood whose closure is compact.
4. X is *metrizable* if τ is compatible with some metric d .
5. X is an *F-space* if its topology τ is induced by a complete invariant metric d . (Compare Section 1.25.)
6. X is a *Fréchet space* if X is a locally convex F -space.
7. X is *normable* if a norm exists on X such that the metric induced by the norm is compatible with τ .
8. *Normed spaces* and *Banach spaces* have already been defined (Section 1.2).
9. X has the *Heine-Borel property* if every closed and bounded subset of X is compact.

Remark. “We shall now define another locally convex topology τ on $\mathcal{D}$ in which Cauchy sequences do converge. The fact that this τ is not metrizable is only a minor inconvenience, as we shall see.”–Rudin

Remark. “We now define a topology on $C^\infty(\Omega)$ which makes $C^\infty(\Omega)$ into a Fréchet space with the Heine-Borel property, such that $\mathcal{D}_K$ is a closed subspace of $C^\infty(\Omega)$ whenever $K \subseteq \Omega$.”–Rudin

Theorem 9. $\mathcal{S}$ is a Fréchet space.

Example 1 (Differentiation of sequences and series). The rather innocuous statement on the sequential continuity of differentiation in $\mathcal{D}'(\Omega)$ (Proposition 3.1.2) is a first indication on the differences between the rules of calculation in classical functions versus distributions. Indeed, it implies that any sequence (or, which amounts to the same, series) converging in $\mathcal{D}'$ can be differentiated term-wise, in stark contrast to classical analysis.

Lemma 2. Let M denote the subspace of $D(\Omega_1 \times \Omega_2)$ defined by the linear span of the set

$$M_0 := \{\phi \otimes \psi \mid \phi \in D(\Omega_1), \psi \in D(\Omega_2)\}.$$

Then M is dense in $D(\Omega_1 \times \Omega_2)$.

7.2.2 What are distributions?

Theorem 10. Let $T \in D'(\mathbb{R}^n)$ and $\Omega \subseteq \mathbb{R}^n$ be open and bounded. Then there exists $f \in C^\infty(\mathbb{R}^n)$ and $\alpha \in \mathbb{N}_0^n$ such that

$$T|_\Omega = \partial^\alpha (f|_\Omega).$$

Thus, locally every distribution is the (distributional) derivative of a continuous function.

Theorem 11. Let $T \in E'(\mathbb{R}^n)$ and U be an open neighborhood of $\text{supp}(T)$. Then we can find $m \in \mathbb{N}$ and functions $f_\beta \in C(\mathbb{R}^n)$ ($|\beta| \leq m$) with $\text{supp}(f_\beta) \subseteq U$ such that

$$T = \sum_{|\beta| \leq m} \partial^\beta f_\beta.$$

In other words, every compactly supported distribution can be (globally) represented by a finite sum of (distributional) derivatives of continuous functions.

Theorem 12. Any temperate distribution is a derivative of some polynomially bounded continuous function.

7.2.3 Hypoellipticity

Definition 4. 1. The singular support of a distribution $T \in D'(\Omega)$ is the complement of the largest open set $U \subseteq \Omega$ such that $T|_U \in C^\infty(U)$.

2. A differential operator $P(\partial) = \sum_{|\alpha| \leq m} a_\alpha \partial^\alpha$, $a_\alpha \in \mathbb{C}$, is called hypoelliptic if

$$\text{sing supp}(P(\partial)T) = \text{sing supp}(T)$$

for each $T \in D'(\mathbb{R}^n)$.

Definition 5. Let $P(D) = \sum_{|\alpha| \leq m} a_\alpha D^\alpha$ be a linear partial differential operator with constant coefficients. Then $P(\xi) := \sum_{|\alpha| \leq m} a_\alpha \xi^\alpha$ is called the symbol of P and the homogeneous polynomial

$$\sigma_P(\xi) := \sum_{|\alpha|=m} a_\alpha \xi^\alpha := P_m(\xi)$$

is called the principal symbol of P . The $P_m(D)$ is called the principal part of P . P is called elliptic if $\sigma_P(\xi) \neq 0$ for each $\xi \neq 0$.

Theorem 13. Let P be an elliptic PDO with constant coefficients. Then for all $T \in D'(\Omega)$ we have

$$\text{sing supp } T = \text{sing supp } P(T).$$

In other words, any elliptic operator is hypoelliptic.

Theorem 14. Let $P(D)$ be a linear PDO with constant coefficients. Then the following are equivalent:

1. $P(D)$ is hypoelliptic.
2. There exists a fundamental solution E of $P(D)$ with $\text{sing supp } E = \{0\}$.

Example 2. (iv) The main application of microlocal analysis is the study of regularity properties of partial differential equations. If P is a PDO of order m with C^∞ coefficients and principal part P_m , the characteristic set of P is defined as

$$\text{Char}(P) := \{(x, \xi) \in \mathbb{R}^n \times (\mathbb{R}^n \setminus \{0\}) \mid P_m(x, \xi) = 0\}.$$

Then for any $T \in \mathcal{D}'(\mathbb{R}^n)$, we have the following fundamental result on the propagation of singularities

$$\text{WF}(T) \subseteq \text{Char}(P) \cup \text{WF}(P(T))$$

Thus, if T is a solution to the PDE $P(T) = f$, then the singularities of T are contained in the set $\text{Char}(P) \cup \text{WF}(f)$. Note that, if P is elliptic then $\text{Char}(P) = \emptyset$. In this case, due to Proposition 6.4.11 we obtain that $\text{WF}(T) = \text{WF}(P(T))$, i.e., the singularity set is exactly preserved. In particular, by applying Pr_1 we confirm the elliptic regularity Theorem 6.2.4, i.e., $\text{singsupp}(P(T)) = \text{singsupp}(T)$.

8 April 17

8.1 Papers

- <https://www.mathnet.ru/>

9 April 23, Polynomial growth harmonic functions

Preparing for chap 28, from reviewing chap 27, Linear growth harmonic functions.

9.1 Papers

- Permutation and combination formula: [oiwiki](#), also [amazon](#)
- Interesting about spinors: [arxiv](#)

9.2 Mathematics

Lemma 3 (Li). *Let M^m be a complete manifold with nonnegative Ricci curvature. Suppose f is a bounded subharmonic function defined on M , then*

$$\lim_{\rho \rightarrow \infty} V_p^{-1}(\rho) \int_{B_p(\rho)} f(x) dx = \sup_M f.$$

